Zhuosheng Liu, Ph.D.

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Applied AI/ML • AI for Science • Bioinformatics • Microbial Food Safety

Research Summary

Researcher combining domain knowledge in microbial food safety with bioinformatics and applied AI/ML. Work spans interpretable pathogen genomics, protein representation learning, generative-AI evaluation, and large-scale recommendation systems. Experience connects experimental science, model development, and evaluation in research and production settings.

Education

University of California, Davis

Ph.D. *2020–2026*

Research: microbial food safety, bioinformatics, and machine learning. Advisor: Luxin Wang. GPA: 3.97/4.00.

University of California, Davis

Master's degree, Computer Science *2023–2025*

GPA: 3.91/4.00.

Columbia University

Master's degree, Biotechnology *2019–2020*

Thesis-based program. GPA: 3.84/4.00.

University of California, Davis

Bachelor's degree, Food Science and Technology *2019*

GPA: 3.59/4.00; major GPA: 3.62/4.00.

Highlighted Research Directions

Interpretable machine learning for microbial genomics

Combined pangenome analysis and ML/deep learning to distinguish clinical from environmental Vibrio parahaemolyticus isolates. Led investigation, data curation, methodology, formal analysis, and drafting; used model interpretation to prioritize candidate genes for biological investigation.

Protein representation learning and structural analysis

Combined curated TDH/TRH sequences, ESM2 embeddings, and AlphaFold2 predictions to investigate toxin divergence. Used clustering and sequence–structure analysis to develop hypotheses relevant to experimental strain selection.

Pathogen transcriptomics and aquaculture microbiomes

Investigated Vibrio survival and gene-expression responses in seawater at shellfish harvesting temperatures. Contributed to studies of florfenicol-associated changes in catfish microbiota and antimicrobial-resistance questions in aquaculture.

Microbial safety of alternative proteins, produce, and dried fruit

Studied pathogen and spoilage-organism behavior in plant-based meat analogues and ground beef; evaluated microbial control during dried-fruit preparation and storage. Contributed to citrus and mushroom food-safety studies.

Editorial Appointment

Microbiome Data Science, Frontiers in Microbiology

Review Editor *September 2026–present*

Appointed to the section editorial board to review manuscripts in computational, statistical, and informatics approaches to microbiome research.

Industry Experience

Meta Platforms

Ph.D. Intern, Machine Learning *June 2026–present*

Developed LLM-based music and creator recommendation systems for content creation. Built pipelines for semantic tokenization, supervised fine-tuning, preference optimization, and online evaluation; **designed Atomic Creator Tokens for creator recommendation**.

Capital One, McLean, VA

Ph.D. Data Science Intern, AI Foundation *June–August 2025*

Investigated graph representation learning using data from **approximately two million customers** to predict shared accounts and support customer modeling and fraud-risk analysis.

Metagenomi, Emeryville, CA

Machine Learning Intern *June–September 2024*

Developed preprocessing, feature engineering, and ML/deep-learning models for CRISPR nuclease indel profiles and gene-editing outcomes using internal screening data.

Publications

† Equal contribution / co-first author. Publication status is indicated for each entry; published versions and earlier preprints are listed together. Impact Factor: the journal's highest Clarivate Journal Citation Reports value, with its data year.

1. [Preprint] **Liu, Zhuosheng**, Zhou, Y., Liu, C., Brown, C. T., & Wang, L. (2026). How thermostable direct haemolysin (TDH) diverges from TDH-related haemolysin (TRH)? Reassessing haemolysins in Vibrio parahaemolyticus through functional and structural representation learning. bioRxiv; posted September 3, 2026. doi: 10.64898/2026.09.02.748974

2. [Journal article] **Liu, Zhuosheng**, Li, Z., Zhang, J., Brown, C. T., & Wang, L. (2026). Interpretable learning algorithms enable pathogenic potential assessment and virulence-associated gene discovery of Vibrio parahaemolyticus. **Frontiers in Microbiology**, 17, 1832130. **Impact Factor: 6.064** (2021). doi: 10.3389/fmicb.2026.1832130 Earlier preprint: doi: 10.1101/2025.04.08.647818

3. [Conference paper] Xiang, X., Liu, X., Li, Z., **Liu, Zhuosheng**, & Zhang, J. (2026). AIGVE-Tool: AI-Generated Video Evaluation Toolkit with Multifaceted Benchmark. **Proceedings of the 21st International Conference on Computer Vision Theory and Applications (VISAPP)**, Vol. 1, 346–357. doi: 10.5220/0014252500004084 Earlier preprint: arxiv.org/abs/2503.14064

4. [Manuscript under review] Beyond Semantic IDs: Atomic Creator Tokens for Large-Scale Generative Recommendation (2026). **Zhuosheng Liu: first author**.

5. [Research manuscript] From Listening to Creating: Preference Optimization for LLM-Native Music Recommendation (2026). Zhuosheng Liu: third author.

6. [Journal article] Tu, T.†, **Liu, Zhuosheng**†, Li, X., Guo, C., Chen, Z., Wang, H., & Wang, L. (2025). Survival of Listeria monocytogenes on growing and harvested Trumpet Royale (Pleurotus eryngii), Alba Clamshell (Hypsizygus tessellatus), and Brown Clamshell (Hypsizygus tessellatus) mushrooms. **Food Microbiology**, 130, 104778. **Impact Factor: 6.374** (2021). doi: 10.1016/j.fm.2025.104778

7. [Journal article] Wang, H., Sheng, L., Yazdi, Z., Li, X., **Liu, Zhuosheng**, Canakapalli, S., Zhou, Y., Liao, C., Emami, S., Kelly, A. M., Roy, L. A., Soto, E., & Wang, L. (2025). The impact of florfenicol treatment on the microbial populations present in the gill, intestine, and skin of channel catfish (Ictalurus punctatus). **Animal Microbiome**, 7, 68. **Impact Factor: 4.5** (2025). doi: 10.1186/s42523-025-00433-9

8. [Preprint] Liu, X., Xiang, X., Li, Z., Wang, Y., Li, Z., **Liu, Zhuosheng**, Zhang, W., Ye, W., & Zhang, J. (2024; revised 2026). A Survey of AI-Generated Video Evaluation. arXiv:2410.19884. doi: 10.48550/arXiv.2410.19884

9. [Journal article] **Liu, Zhuosheng**, Zhou, Y., Wang, H., Liu, C., & Wang, L. (2024). Recent advances in understanding the fitness and survival mechanisms of Vibrio parahaemolyticus. **International Journal of Food Microbiology**, 417, 110691. **Impact Factor: 5.911** (2021). doi: 10.1016/j.ijfoodmicro.2024.110691

10. [Journal article] Wang, H., Sheng, L., **Liu, Zhuosheng**, Li, X., Harris, L. J., & Wang, L. (2024). Reduction Foodborne Pathogens and Surrogate Microorganism on Citrus Fruits after Lab- and Pilot-

scale Finishing Wax Application. **Journal of Food Protection**, 87, 100255. **Impact Factor: 3.7** (2025). doi: 10.1016/j.jfp.2024.100255

11. [Journal article] **Liu, Zhuosheng**†, Liao, C.†, & Wang, L. (2023). Fitness and transcriptomic analysis of pathogenic Vibrio parahaemolyticus in seawater at different shellfish harvesting temperatures. **Microbiology Spectrum**, 11(6), e02783-23. **Impact Factor: 4.1** (2025). doi: 10.1128/spectrum.02783-23 Earlier preprint: doi: 10.1101/2023.05.03.539256

12. [Journal article] **Liu, Zhuosheng**, Shaposhnikov, M., Zhuang, S., Tu, T., Wang, H., & Wang, L. (2023). Growth and survival of common spoilage and pathogenic bacteria in ground beef and plant-based meat analogues. **Food Research International**, 164, 112408. **Impact Factor: 8.8** (2025). doi: 10.1016/j.foodres.2022.112408

13. [Journal article] **Liu, Zhuosheng**, Sheng, L., Canakapalli, S. S., & Wang, L. (2022). Evaluation of the microbial control efficacies of commonly used home-drying and storage practices of dried peaches. **LWT**, 165, 113734. **Impact Factor: 7.1** (2025). doi: 10.1016/j.lwt.2022.113734

14. [Journal article] **Liu, Zhuosheng**, Liao, C., Golson, K., Phillips, S., & Wang, L. (2021). Survival of common foodborne pathogens on dried apricots made with and without sulfur dioxide treatment. **Food Control**, 121, 107569. **Impact Factor: 7.0** (2025). doi: 10.1016/j.foodcont.2020.107569

15. [Technical report] Li, Z., **Liu, Zhuosheng**, & Zhang, J. Parameter-Efficient Fine-Tuning for Vision-Language Models. Technical report

Research Affiliations

Information Fusion and Mining Laboratory, UC Davis

Research contributor *May 2023–present*

Research on parameter-efficient fine-tuning for vision-language models and evaluation of AI-generated video.

Luxin Wang Laboratory, UC Davis

Doctoral research *2020–2026*

Experimental and computational investigations of pathogen survival, food-safety risks, microbial genomes, and microbiomes.

Technical Expertise

AI/ML: PyTorch, TensorFlow, scikit-learn, Hugging Face; LLM fine-tuning, LoRA, PEFT, DPO, and GRPO, multimodal representation learning, graph learning, generative recommendation, and model evaluation.

Programming and systems: Python, R, C/C++, Java, SQL, Scala, Perl, Bash; NumPy, pandas, Spark/PySpark, PostgreSQL, Linux, Git, Docker, AWS, FSDP, and vLLM.

Bioinformatics & Experimental Methods

Snakemake workflows; RNA-seq, 16S rRNA sequencing, whole-genome and pangenome analysis; Biopython; ESM2 protein embeddings, UMAP/t-SNE, and AlphaFold2 structure prediction.

NGS library preparation, qPCR, PCR, microbial culture, pathogen challenge studies, and food-safety experiments.

Research Recognition

Sweden's Food Agency discussed the plant-based-meat study and **reproduced measured product pH values in its 2025 microbiological risk assessment** of plant-based alternatives.

A 2024 **Eurosurveillance** report on a Finnish hospital outbreak **cited the plant-based-meat study as evidence** that pea-protein alternatives can support Listeria growth.

Peer-Review Service

Journal reviewing

Frontiers in Microbiology — 2025–2026.

Frontiers in Marine Science — 2025.

Journal of Food Science — 2026.

IEEE Transactions on Neural Networks and Learning Systems — 2025.

IEEE Transactions on Multimedia — 2025.

IEEE Journal of Biomedical and Health Informatics — 2025.

IEEE Data Descriptions — 2025.

ACM Transactions on Computing for Healthcare — 2025.

ACM Computing Surveys — 2026.

ACM Transactions on Sensor Networks — 2026.

Journal of Medical Imaging — 2025.

Conference, track, and workshop reviewing

ICLR Workshop on Foundation Models for Science (FM4Science), 2026 — **six completed reviews**.

ICLR-associated reviewing — 2024.

NeurIPS workshops: Imagenomics and AI4DS — 2025.

NeurIPS reviewing — 2026.

KDD AI4Science — 2026.

KDD 5th Workshop on Ethical Artificial Intelligence.

KDD 5th Workshop on Uncertainty Reasoning and Quantification in Decision Making.

ACM CAIS AgentSkills Workshop — 2026.

IEEE Cloud Summit.

International Conference on Artificial Neural Networks.

IFT FIRST Technical Research Paper reviewing — 2023.

Selected publicly credited reviews

Frontiers in Microbiology: "Preliminary study of the growth variability of Vibrio parahaemolyticus in simulated gastric digestion fluids" (published 2026). **Named reviewer.** doi: 10.3389/fmicb.2026.1753353

Frontiers in Marine Science: "Proteomics and metabolic profiling of Vibrio parahaemolyticus at three different temperatures" (published 2025). **Named reviewer.** doi: 10.3389/fmars.2025.1689231

Honors & Awards

Margrit Mondavi Student Support Fellowship, UC Davis	2025
LUGG Graduate Student Award, UC Davis	2024
Jastro-Shields Research Award / Peter J. Shields and Henry A. Jastro Research Award, UC Davis	2021–2024
First Place, NCIFT Student Poster Competition	2023
James and Marilyn Lugg Graduate Student Award, UC Davis	2022, 2023
Fresh Express Incorporated Graduate Student Support Fund, UC Davis	2022, 2023
Victor Chu Endowment, UC Davis	2022
Bor Luh Scholarship; George Stewart Memorial Award, UC Davis	2021
Doctoral Student Fellowship, UC Davis	2020
Departmental Master's Student Scholarship, Columbia University	2019
Third Place, CAFPNA Student Presentation Competition	2019
Marie and William Cruess Scholarship, NCIFT	2019
Departmental Citation for Outstanding Performance; Undergraduate Research Travel Award, UC Davis	2019
Provost's Undergraduate Research Fellowship, UC Davis	2018

Teaching Experience

UC Davis

Graduate Teaching Assistant, Microbiology / Food Science — *From March 2021*

Supported FST 104L and MIC 103L laboratory instruction; held office hours and graded work for FST 10 (approximately 300 students); led discussions for FST 100B.

UC Davis

Graduate Teaching Assistant, Computer Science — *From September 2024*

Supported ECS 122A (Algorithm Design and Analysis), serving approximately 600 students.

UC Davis

Undergraduate Learning Assistant *August 2017–April 2019*

Tutored students in MIC 103 laboratory coursework and the CHE 2 general chemistry series.

Selected Presentations

2024 | **Liu, Zhuosheng**, et al. Pathogenicity prediction of Vibrio parahaemolyticus using pangenome data and machine learning. IAFP Annual Meeting, Long Beach, CA.

2023 | Wang, H., et al., including **Liu, Zhuosheng**. Fate of foodborne pathogens on lemons after lab- and pilot-scale finishing wax application. IAFP Annual Meeting, Toronto, Canada.

2023 | **Liu, Zhuosheng**, Liao, C., & Wang, L. Fitness and transcriptomic analysis of Vibrio parahaemolyticus at oyster harvesting temperatures. Aquaculture America, New Orleans, LA.

2022 | **Liu, Zhuosheng**, Liao, C., & Wang, L. Vibrio fitness and transcriptomic analysis. Pacific Fisheries Technologists Conference, Newport, OR; and IAFP Annual Meeting, Pittsburgh, PA.

2022 | **Liu, Zhuosheng**, et al. Evaluation of the microbial quality of plant-based meat analogues. IAFP Annual Meeting, Pittsburgh, PA.

2022 | Wang, H., et al., including **Liu, Zhuosheng**. Impact of florfenicol treatment on microbial populations associated with live catfish. IAFP Annual Meeting, Pittsburgh, PA.

2019 | **Liu, Zhuosheng**, Liao, C., & Wang, L. Behavior of foodborne pathogens on dried apricots made with and without sulfur dioxide. IAFP Annual Meeting, Louisville, KY.